ALEX CHEVRIER
Low-Latency C++ Systems Engineer
Hong Kong | +852 55189053 | chevrier.alex@gmail.com |
linkedin.com/in/alex-chevrier-323365b4 | github.com/alchevrier

SKILLS

Languages: C++23, Java 25, SQL
Low-latency: lock-free SPSC/MPSC, seqlock, mmap, DPDK, memory ordering, pthread_setaffinity_np, perf stat, GBench/GTest, CMake/Meson, Conan
Systems: Linux/GCC, Kafka, binary TCP/NIO, Docker, AWS

PROJECTS

Low-Latency Feed Handler (C++23) | github.com/alchevrier/low-latency-feed-handler
• NASDAQ ITCH 5.0 zero-copy parser → MPSC → SOA order book; DPDK pcap PMD ingestion layer. Zero heap end-to-end.
• Real NASDAQ dataset (11 GB, 163M messages, 8,669 symbols); P99.9 274 ns parse + book insert; 97.5% branch accuracy via perf stat.

Low-Latency Order Book (C++23) | github.com/alchevrier/low-latency-order-book
• N-SPSC MPSC, SOA order book (sorted int64_t tick arrays), seqlock sync, pinned threads.
• Median 2.5 ns / p99 4 ns isolated; <1 ns seqlock overhead under concurrent writes. Zero heap (massif). 10 ADRs.

Low-Latency Log Engine (C++23) | github.com/alchevrier/low-latency-log-engine
• mmap-backed SPSC log; median 4.3 ns / p99 9.5 ns. Zero heap across 1M iterations. C++23 concepts. 13 ADRs.

EXPERIENCE

Morgan Stanley, Hong Kong — Software Engineer (Contract) Nov 2025–Present
• Designed greenfield messaging service; reverse-engineered Python batch system into streaming Kafka pipeline.
• Maintain production Kafka infrastructure across isolated data centres.

We+, Hong Kong — Software Engineer Sept 2023–Nov 2025
• Migrated highest number of microservices to Java 21 across French investment bank FX platform.

LandyTech, London — Software Engineer Oct 2019–Aug 2023
• Monolith-to-microservices migration; automated risk data pipeline.

Redbite Solutions, Cambridge — Software Engineer Nov 2016–Oct 2019
• Built web application from scratch; implemented OAuth2 authentication layer.

EDUCATION

ENSEIRB-MATMECA, Bordeaux — M.Sc. Computer Science 2013–2016
Admitted by competitive national examination (Grande École entrance).